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Session

'A'

Course

Web System
Technology (WST)
— CS-305 — IV (E) BSCS

(JS-NOTES)

Instructor

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Variables & Datatypes

Definition

"Variables are containers used to store data in Javascript."

Javascript Provides three ways to declare variables:

- var
- let
- Const

Purpose

Variables help developers store and manage application data such as:

- usernames
- login information
- scores
- Product Prices
- settings

Syntax

```
let age = 20;  
const name = "Sehrish";  
var city = "Gujrat";
```

Example

```
let studentName = "Sehrish";  
let marks = 95;  
let isPassed = true;  
console.log(studentName);  
console.log(marks);  
console.log(isPassed);
```

Var vs let vs const

Keyword	Scope	Reassign	Redeclare
Var	Function	Yes	Yes
let	Block	Yes	No
const	Block	No	No

* Data Types

(i) Primitive Data Types

- String
- Number
- Boolean

- Undefined
- Null

ii) Non-Primitive

- Object
- Array

2. Type Checking

```
let name = "Sehrish";  
console.log(typeof name);
```

3. Type Conversion

1. Implicit Conversion

```
console.log(10 + "5");
```

2. Explicit Conversion

```
Number("100");
```

```
String(50);
```

```
Boolean(1);
```

Real World Use

Variables are used in:

- login systems
- e-commerce websites
- dashboards
- portfolio websites
- forms
- and much more.

Important Points

- `const` can't be reassigned.
- `let` is preferred in modern Javascript.
- `typeof` checks Variable type.
- Arrays and objects are reference types.